EXPERIMENT 2: Write a Google App Engine Program to Generate Nth Odd Numbers and Deploy It to Google Cloud

1. Aim

To develop a Python web application using Google App Engine to generate the first N odd numbers and deploy it to Google Cloud.

2. Procedure

1. Open Google Cloud Console at https://console.cloud.google.com/ and sign in.
2. Create a new Google Cloud project and select it.
3. Enable billing for the project if required by the account or deployment configuration.
4. Install Python and the Google Cloud CLI on the local computer.
5. Open a terminal and initialize the CLI using gcloud init; sign in and select the project.
6. Create a project directory named OddNumbers.
7. Inside the directory, create main.py and enter the Flask application shown below.
8. Create requirements.txt containing the Flask dependency.
9. Create app.yaml and specify the supported App Engine Python runtime.
10. Deploy the application using gcloud app deploy.
11. After deployment completes, open the application using gcloud app browse.
12. Enter a value for N and verify the generated odd numbers.

3. Program

File – main.py:

```
from flask import Flask, request

app = Flask(__name__)

@app.route("/")
def odd_numbers():
    n = request.args.get("n")

    if n is None:
        return '''
        <h2>Generate Odd Numbers</h2>
        <form>
          Enter N:
          <input type="number" name="n" min="1" required>
          <input type="submit" value="Generate">
        </form>
        '''

    try:
        n = int(n)
        if n < 1 or n > 1000:
            return "Enter a number between 1 and 1000"

        result = [2 * i + 1 for i in range(n)]
        return f"<h2>First {n} Odd Numbers</h2>{result}"
    except ValueError:
        return "Enter a valid integer"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=8080)
```

File – requirements.txt:

```
Flask
```

File – app.yaml:

```
runtime: python313
```

Deployment commands:

```
gcloud init
gcloud app deploy
gcloud app browse
```

4. Output of Program

```
Browser input: N = 5

First 5 Odd Numbers:
[1, 3, 5, 7, 9]
```

Note: Keep main.py, requirements.txt and app.yaml in the same application directory. The runtime must be supported by the installed App Engine environment.

5. Result

Thus, the Python web application to generate the first N odd numbers was developed and deployed using Google App Engine.

EXPERIMENT 2: Write a Google App Engine Program to Generate Nth Odd Numbers and Deploy It to Google Cloud

1. Aim

To develop a Python-based web application using the Flask framework that generates the first N odd numbers and deploy the application on Google App Engine, a Platform as a Service (PaaS) offered by Google Cloud.

2. Procedure

1. Open a web browser and navigate to the Google Cloud Console. 2. Sign in using a valid Google account and create a new Google Cloud project. 3. Select the newly created project and enable the required Google Cloud services. 4. Enable billing if it is required for the selected deployment configuration. 5. Install Python and the Google Cloud CLI on the local computer. 6. Open the terminal or command prompt and execute gcloud init to initialize the Google Cloud CLI. 7. Authenticate using the Google account and select the appropriate Google Cloud project. 8. Create a new application directory named OddNumbers to store the project files. 9. Inside the directory, create a Python file named main.py. 10. Import the Flask module and create a Flask application instance. 11. Define a route that accepts the value N from the user's browser input. 12. Implement the logic to generate odd numbers using the mathematical expression 2 × i + 1. 13. Add input validation to ensure that N is a valid positive integer. 14. Create a file named requirements.txt and include Flask as the application dependency. 15. Create an app.yaml file and specify the supported Python runtime. 16. Open the terminal inside the application directory and execute gcloud app deploy. 17. Wait until the deployment process completes successfully. 18. Execute gcloud app browse to open the deployed application in the browser. 19. Enter a sample value such as 5 and verify that the first five odd numbers are displayed correctly.

4. Output of Program

Input: N = 5 Expected Browser Output: First 5 Odd Numbers [1, 3, 5, 7, 9] Explanation: The application accepts the number of terms from the user and generates the first N odd numbers. The odd numbers are calculated using the formula: [Odd(i)=2i+1] The application is hosted on Google App Engine and can be accessed through its deployed URL.

5. Result

Thus, a Python web application was successfully developed using Flask to generate the first N odd numbers. The application was deployed on Google App Engine and tested through a web browser, producing the expected output.